

5	(a)	(i)	Simple harmonic motion is defined as the motion of a particle about a fixed point such that its acceleration a is proportional to its displacement x from a fixed point, and is directed towards the fixed point.	
		(ii)	use of $a = -\omega^2 x$ clear either $\omega = \sqrt{(2k/m)}$ or $\omega^2 = (2k/m)$ $\omega = 2\pi f$ $f = (1/2\pi)\sqrt{(2 \times 300)/0.240}$ $= 7.96 \approx 8 \text{ Hz}$	
	(b)	(i)	1. resonance 2. 8 Hz	
		(ii)	There is maximum energy transfer from the driver to the oscillating mass during resonance.	
		(iii)	(increase amount of) damping without altering (k or) m ...(some indirect reference is acceptable) sensible suggestion.	

6	(a)		Into the page / into the (plane of) paper.	
---	-----	--	--	--